

Clinically-Grounded Machine Learning

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The Question

What is a simple and efficient way to incorporate clinical knowledge into ML models?

Motivation

- ▶ healthcare decisions: high stakes, limited resources
- ▶ trust requires **interpretability and accuracy**
- ▶ regulations (EU AI Act, FDA) demand explainability

Right to Explanation of Individual Decision-Making

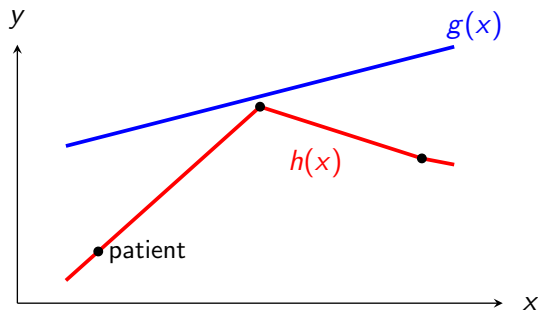
According to article 86 of the EU AI Act,

“Any affected person . . . shall have the right to obtain from the deployer clear and meaningful explanations of the role of the AI system in the decision-making procedure and the main elements of the decision taken.”



**Require ML predictions to be consistent
for clinically similar patients.**

Core Idea - Scatterplot



Use simple model $g(x)$, which captures **clinical similarity**, to guide training of **high-dim. model $h(x)$** .

Clinical Similarity

- ▶ use key predictive features
- ▶ based on staging systems
- ▶ ...

Single Most Predictive Risk Factors for Diseases

- ▶ stroke → hypertension¹
- ▶ type 2 diabetes → obesity (high BMI)²
- ▶ lung cancer → cigarette smoking³

¹Gaciong Z, et.al., Blood Pressure Control and Primary Prevention of Stroke: Summary of the Recent Clinical Trial Data and Meta-Analyses. *Curr Hypertens Rep.* 2013;15(6):559–574. PMCID: PMC3838588, PMID: 24158454

²Haghighatdoost F, et.al., Are body mass index and waist circumference significant predictors of diabetes and prediabetes risk: Results from a population based cohort study. *World J Diabetes.* 2017;8(7):365–373. PMCID: PMC5507834, PMID: 28751960

³U.S. Center for Disease Control and Prevention. Lung Cancer Risk Factors. <https://www.cdc.gov/lung-cancer/risk-factors/index.html>

Common Staging / Classification Systems

- ▶ stroke → NIH stroke scale (NIHSS)⁴
- ▶ type 2 diabetes → ADA diagnostic categories (normal, prediabetes, diabetes)⁵
- ▶ lung cancer → TNM staging (AJCC/UICC)⁶

⁴Brott T, et al. Measurements of acute cerebral infarction: a clinical examination scale. *Stroke*. 1989;20(7):864–870. doi:10.1161/01.STR.20.7.864

⁵American Diabetes Association. Standards of Care in Diabetes—2025. *Diabetes Care*. 2025;48(Suppl. 1):S1–S200. doi:10.2337/dc25-Sintroduction

⁶Amin MB, et al. *AJCC Cancer Staging Manual*. 8th ed. Springer; 2017. doi:10.1007/978-3-319-40618-3

Example: Key Predictive Feature

atherosclerosis

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Lp(a) testing for the primary prevention of cardiovascular disease in high-income countries: a cost-effectiveness analysis

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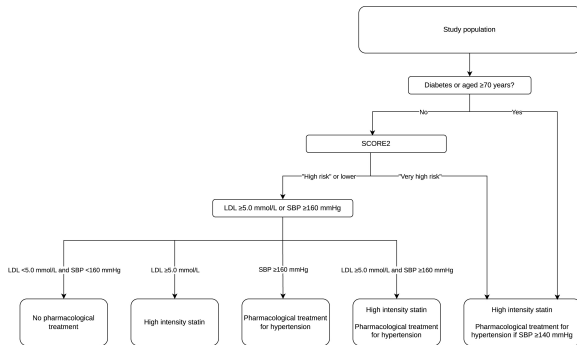
Highlights

Show Outline

- First cost-effectiveness study of routine Lp(a) testing for primary CVD prevention across multiple high-income countries.
- Lp(a) testing in primary prevention may reclassify CVD risk and be cost-saving from both healthcare and societal

Figure

Example: Staging System



Reproduced from Morton, J. I. *et al.*, "Lp(a) testing for the primary prevention of cardiovascular disease in high-income countries: a cost-effectiveness analysis," *Atherosclerosis*, Vol. 0, Issue 0, 120447, 2025.

<https://doi.org/10.1016/j.atherosclerosis.2025.120447>. Licensed under CC BY 4.0.

Interpretable ERM

learn predictor $\hat{h} \in \mathcal{H}$ via

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \left(\frac{1}{m} \sum_{i=1}^m \ell(h(\mathbf{x}^{(i)}), y^{(i)}) + \alpha \mathcal{R}(h) \right)$$

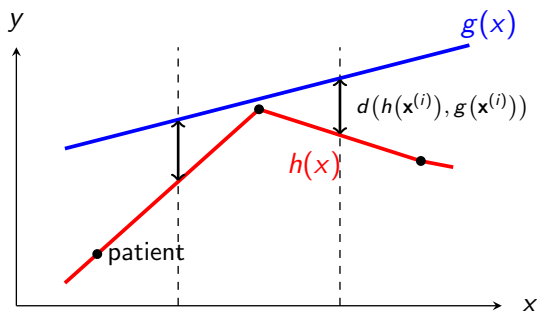
where $\mathcal{R}(h)$ penalizes variability across clinically similar patients

- ▶ $\mathcal{R}(h)$ measures (lack of) interpretability of h
- ▶ parameter α controls level of interpretability
- ▶ two constructions for clinically-grounded regularizer $\mathcal{R}(h)$

Construction I

- ▶ choose key features x_j for $j \subseteq \{1, \dots, n\}$
- ▶ train low-dim. model $g(x)$ using key features
- ▶ compute discrepancy between $h(x)$ and $g(x)$ on test-set,

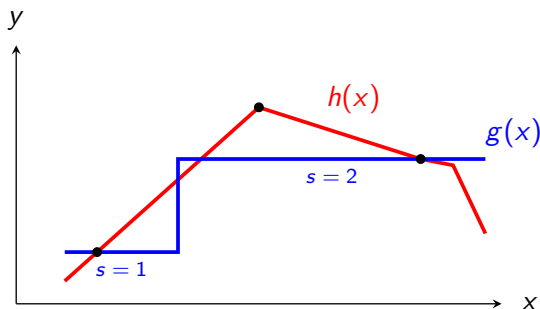
$$\mathcal{R}(h) = (1/m_t) \sum_{i=1}^{m_t} d(h(\mathbf{x}^{(i)}), g(\mathbf{x}^{(i)}))$$



Construction II

- ▶ subset $\mathcal{D}^{(s)}$ of patients in stage/class/scale $s = 1, \dots, S$
- ▶ average stage-wise variations,

$$\mathcal{R}(h) = \sum_{s=1}^S \frac{1}{|\mathcal{D}^{(s)}|} \sum_{\mathbf{x} \in \mathcal{D}^{(s)}} (h(\mathbf{x}) - \mu^{(s)})^2, \quad \mu^{(s)} = \frac{1}{|\mathcal{D}^{(s)}|} \sum_{\mathbf{x} \in \mathcal{D}^{(s)}} h(\mathbf{x}).$$



Case Study: Cancer Survival Prediction

- ▶ data: multiple myeloma patients at HUS
- ▶ patients with features: age + blood biomarkers
- ▶ label: 5-year survival outcome

Results

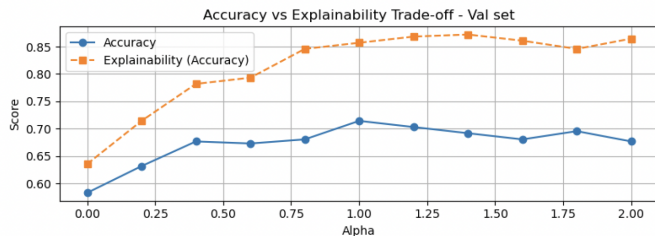


Figure 29: Trade-off between accuracy and explainability for ANN models with respect to the hyperparameter alpha (α) for the validation set. Explainability is measured as the accuracy between the ANN and the auxiliary model.

Reproduced from S. Rachidi, “Explainable machine learning in cancer survival prediction: A case study of myeloma patients”, Master thesis, Aalto University, 2025.

Takeaway

- ▶ we capture clinical knowledge by a regularizer
- ▶ regularizer can be interpreted as measure of interpretability
- ▶ requiring interpretability is a form of regularization
- ▶ higher interpretability can result in higher accuracy!

References

- ▶ S. Rachidi, “Explainable machine learning in cancer survival prediction: A case study of myeloma patients”, Master thesis, Aalto University, 2025.
- ▶ Zhang, L., Karakasidis, G., Odnoblyudova, A. et al. Explainable empirical risk minimization. Neural Comput & Applic 36, 3983–3996 (2024).
<https://doi.org/10.1007/s00521-023-09269-3>